Zhijie Zheng

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EDUCATION

University of California, Davis — Davis, CA
Ph.D. in Computer Science (Advisor: Dr. Jiawei Zhang) | GPA: 3.9/4.0 — September 2025 – June 2030 (Expected)

Sun Yat-sen University — Guangzhou, China
M.S. in Electronic Science | GPA: 3.9/4.0 — June 2025

Sun Yat-sen University — Guangzhou, China
B.S. in Electronic Science | GPA: 3.9/4.0 — June 2022

RESEARCH EXPERIENCE

RDGSplat: Render-Dedicated Geometry for Novel View Synthesis — UC Davis
Zheng et al. *arXiv preprint, 2026.* Paper — July 2026 – Present

- Developed a parallel decoder branch predicting additional 3D Gaussians for novel-view rendering on a frozen 1.4B 3D foundation model, with a target-pose-conditioned adapter and photometric-only supervision.
- Preserved depth and pose predictions, unlike backbone fine-tuning, with only 205M (14.7%) trainable parameters.
- Improved RE10K PSNR from 20.9 to **25.1 dB** and zero-shot PSNR by 2.7–3.8 dB on ACID, DL3DV, and DTU, outperforming pixelSplat and MVSplat on RE10K and DTU; gains held across three backbones.

PRISM: Feed-Forward Single-Image 3D Reconstruction via Geometric Warp-Residual Modeling — UC Davis
Zheng et al. *arXiv preprint, 2026.* Paper | GitHub — February 2026 – June 2026

- Built a feed-forward single-image 3D reconstruction pipeline that forward-warps input latents to target views using monocular depth, with a 26M transformer predicting residuals for LYRA's frozen 3DGS decoder.
- Eliminated diffusion sampling at inference, the main bottleneck of diffusion-based methods: geometric warping covers up to 90% of the target view, so diffusion supervision is needed only for training on 401 synthetic scenes.
- Cut per-scene inference time by **277×** over diffusion-based LYRA (2.8 h to 36 s) while staying within 0.8 dB of its PSNR on RE10K and DL3DV; achieved zero-shot generalization to three real-world benchmarks.

TTSA3R: Training-Free Temporal-Spatial Adaptive Persistent State for Streaming 3D Reconstruction — UC Davis
Zheng et al. *WACV 2027 LVR Workshop (Oral).* Paper | GitHub — October 2025 – February 2026

- Proposed a training-free memory update rule for streaming 3D reconstruction that combines temporal state changes and observation–state alignment to compute token-wise update weights.
- Mitigated catastrophic forgetting in long sequences caused by uniform memory updates, without additional training or parameters and while maintaining the base model's speed (18 FPS).
- Reduced ATE by 43% vs. CUT3R on TUM-dynamics, limited reconstruction error growth from 50 to 250 frames to **1.33×** vs. 3.8×, and achieved the lowest pose error on 1,000-frame sequences.

LiDAR 3D Object Detection for Autonomous Driving — SYSU
4 journal publications, including 2 first-author papers (IEEE TMM, IEEE SPL). — August 2022 – December 2023

- Designed a bidirectional attention module to fuse raw-point and range-view features, providing dense range-view context for sparse points and improving Waymo mAP by 4.2 points over IA-SSD (IEEE TMM).
- Proposed a dual-channel PointNet++/Transformer backbone with density-aware center features for local and long-range geometry, improving KITTI cyclist AP by 4.18 over IA-SSD with ~4× fewer parameters than PDV (IEEE SPL).

WORK EXPERIENCE

Beijing Novauto Technology | Research Intern, LiDAR 3D Perception — March 2024 – October 2024
Zheng et al. *PRICAI 2025 (co-first author).* Paper | GitHub

- Developed a multi-task pretraining framework for multi-dataset LiDAR detection, combining masked point reconstruction and occupancy prediction with cross-dataset attention to align dataset-specific features.
- Enabled joint training across Waymo, nuScenes, and KITTI by addressing sensor and scene discrepancies that cause naive multi-dataset training to underperform.
- Outperformed Uni3D (CVPR 2023) by 2.29 mAPH on Waymo, 1.45 BEV mAP on nuScenes, and 2.08 3D mAP on KITTI; improved Waymo mAPH by 8.85 over single-dataset training using **only 1% of Waymo training labels**.

TECHNICAL SKILLS

Research Interests: 3D Reconstruction, Novel View Synthesis, Geometric Representation Learning, 3D Gaussian Splatting
Programming Languages: Python, C/C++, CUDA
Deep Learning & CV Frameworks: PyTorch, OpenPCDet, OpenCV, Open3D, Point Cloud Library (PCL), scikit-learn
Language Skills: Chinese Mandarin (native), English (fluent)

HONORS & AWARDS

GGCS Excellence Fellowship, Graduate Group in Computer Science, UC Davis — 2025, 2026
National Scholarship for Master's Students, Ministry of Education of China — 2023
Second Prize (South China Division), China College IC Competition — 2023
Outstanding Undergraduate Thesis Award, Sun Yat-sen University — 2022